11 Sequences, Series, and Power Series



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11.8 Power Series

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Power Series (1 of 4)

A **power series** is a series of the form

$$1 \quad \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \mathrm{L}$$

where x is a variable and the c_n's are constants called the **coefficients** of the series.

For each number that we substitute for x, the series (1) is a series of constants that we can test for convergence or divergence. A power series may converge for some values of x and diverge for other values of x.

Power Series (2 of 4)

The sum of the series is a function

$$f(x) = c_0 + c_1 x + c_1 x^2 + \mathrm{L} \; + c_n x^n + \mathrm{L}$$

whose domain is the set of all x for which the series converges. Notice that f resembles a polynomial. The only difference is that f has infinitely many terms.

For instance, if we take $c_n = 1$ for all n, the power series becomes the geometric series

$$\mathbf{2} \quad \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \mathrm{L} \; + x^n + \mathrm{L}$$

which converges when $-1 < x < 1$ and diverges when $|x| \geq 1$.

Power Series (3 of 4)

In fact if we put $x = \frac{1}{2}$ in the geometric series (2) we get the convergent series

$$\sum_{n=0}^{\infty}\left(\frac{1}{2}\right)^n = 1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\mathrm{L}$$

but if we put $x = 2$ in (2) we get the divergent series

$$\sum_{n=0}^{\infty}2^n = 1+2+4+8+16+\mathrm{L}$$

Power Series (4 of 4)

More generally, a series of the form

$$\textbf{3}\quad \sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \mathrm{L}$$

is called a **power series in** $(x - a)$ or a **power series centered at** a or a **power series about** a.

Notice that in writing out the term corresponding to $n = 0$ in Equations 1 and 3 we have adopted the convention that $(x-a)^0 = 1$ even when $x = a$.

Notice also that when $x = a$, all of the terms are 0 for $n \geq 1$ and so the power series (3) always converges when $x = a$.

Example 2

For what values of x is the series $\sum_{n=0}^{\infty} n!x^n$ convergent?

Solution:

We use the Ratio Test. Let $a_n = n!x^n$. If $x \neq 0$, we have

$$\begin{aligned}\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty}\left|\frac{(n+1)!x^{n+1}}{n!x^n}\right| \\ &= \lim_{n\to\infty}(n+1)|x| \\ &= \infty\end{aligned}$$

By the Ratio Test, the series diverges when $x \neq 0$. Thus the given series converges only when $x = 0$.

Interval of Convergence

Interval of Convergence (1 of 4)

For the power series that we have looked at so far, the set of values of x for which the series is convergent has always turned out to be an interval.

The following theorem says that this is true in general.

4 Theorem For a power series $\sum_{n=0}^{\infty} c_n (x-a)^n$, there are only three possibilities:

i. The series converges only when $x = a$.

ii. The series converges for all x.

iii. There is a positive number R such that the series converges if $|x-a| < R$ and diverges if $|x-a| > R$.

The number R in case (iii) is called the **radius of convergence** of the power series.

Interval of Convergence (2 of 4)

By convention, the radius of convergence is $R = 0$ in case (i) and $R = \infty$ in case (ii). The **interval of convergence** of a power series is the interval that consists of all values of x for which the series converges.

In case (i) the interval consists of just a single point a.

In case (ii) the interval is $(-\infty, \infty)$.

In case (iii) note that the inequality $|x - a| < R$ can be rewritten as $a - R < x < a + R$.

Interval of Convergence (3 of 4)

When x is an *endpoint* of the interval, that is, $x = a \pm R$, anything can happen—the series might converge at one or both endpoints or it might diverge at both endpoints.

Thus in case (iii) there are four possibilities for the interval of convergence:

$$(a-R,\ a+R)\quad (a-R,\ a+R]\quad [a-R,\ a+R)\quad [a-R,\ a+R]$$

The situation is illustrated in Figure 1.

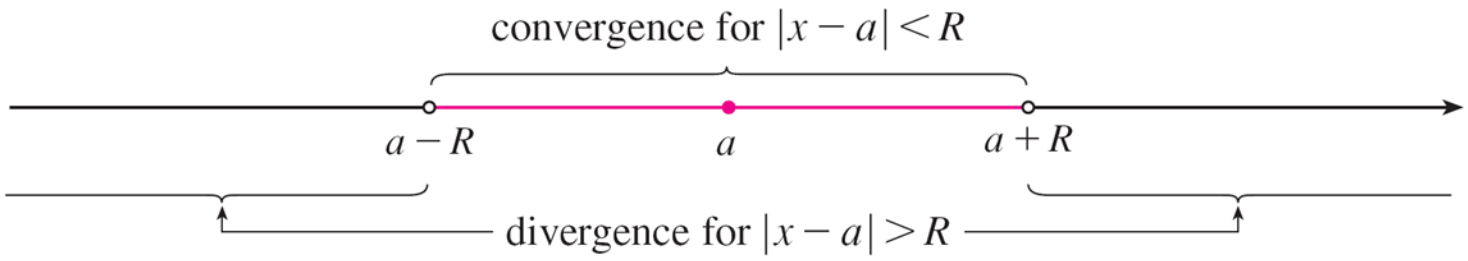


Figure 1

Interval of Convergence (4 of 4)

We summarize here the radius and interval of convergence for each of the examples already considered in this section.

	Series	Radius of convergence	Interval of convergence
Geometric series	$\sum_{n=0}^{\infty} x^n$	$R = 1$	$(-1, 1)$
Example 1	$\sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$	$R = 1$	$[2, 4)$
Example 2	$\sum_{n=0}^{\infty} n!\, x^n$	$R = 0$	$\{0\}$
Example 3	$\sum_{n=0}^{\infty} \frac{x^n}{(2n)!}$	$R = \infty$	$(-\infty, \infty)$